

DAS DIAL PROBLEM STATEMENT 1

Title	DAS Learning Screening Engine
Industry Mentors	Ms Soofrina Mubarak Ms Hakimah Diniyah
Project Context	Dyslexia Association of Singapore (DAS) supports students with dyslexia and other learning differences through specialised literacy intervention programmes. Early identification of literacy difficulties remains a major challenge, as many students are only referred after prolonged academic struggle. Existing screening processes can be resource-intensive and may not provide sufficient granularity for targeted intervention planning. DAS is exploring the development of an AI-assisted Learning Screening Engine that uses short digital cognitive and literacy tasks to identify early indicators of literacy risk. The proposed system aims to support scalable, data-informed screening for students with reading and writing difficulties.
Project Objectives	How can machine learning and educational technology be utilised to: ● Develop a digital literacy screening platform for students ● Analyse learner responses and behavioural patterns ● Predict potential literacy risk indicators using AI/ML models ● Generate interpretable screening summaries for educators ● Support earlier intervention and referral decisions

Project Deliverables	Presentation with demo of prototype Prototype with the following technical requirements: • Web-based application using frontend and backend frameworks • Database for storing anonymised screening results • AI/ML classification model for literacy risk prediction • Dashboard for visualising screening outcomes • Test cases and evaluation metrics Documentation • Architecture diagram and design decisions • Machine learning pipeline and preprocessing workflow • Dataset structure and feature engineering rationale • Project management methodology • User testing findings and evaluation
Main form of expected project deliverables	A web-based AI-assisted literacy screening platform
Crucial properties to test for the delivered software	Functionality, Accuracy, Interpretability, Usability, Security
Additional related resources	www.das.org.sg